



THE FUTURE OF NEURONAL DEATH PREVENTION & NEUROLOGICAL DISORDERS TREATMENTS



Series A Fundraising - Investors Deck



NeurAegis

A late preclinical stage neuropharmaceutical company founded in 2016

Pioneering the development of **first-in-class** compounds, specifically designed to treat major neurological disorders, focusing on **selective Calpain-2 inhibitors**

Our product portfolio addresses significant **unmet medical needs, targeting serious and growing neurological conditions**

THE NEURAEGIS OPPORTUNITY

POTENTIAL TO BE FIRST-IN-CLASS TO PREVENT NEURONAL DEATH

Portfolio of novel, first-in-class protease inhibitors & potentially transformative therapies for multiple neurological disorders with long-term disabling neurodegenerative brain effects

» **Evident unmet medical need**

- » Blockbuster potential as first-in-class assets (Ozimpic-like drug)
- » Large addressable markets **with broad opportunities & target indications**
 - Lead indications are concussion (where there is currently **no FDA approved medication**) and epilepsy
 - **Broad applicability** in other indications, such as Loud Noise-Induced Hearing Loss, Glaucoma, Alzheimer's Disease, and Parkinson's Disease

- » Strong intellectual property portfolio
- » Increased likelihood of clinical success:
 - Well-defined mechanism of action
 - **High protease selectivity**
 - Strong clinical hypothesis
 - **Non-invasive plasma biomarkers** reflecting brain target engagement
- » Experienced, proven leadership team

**RAISING \$10M SERIES A TO FUND TO KEY MILESTONE/ VALUE
INFLECTION POINT OF COMPLETION OF PHASE 1 (IN CONCUSSION/TBI)**

TARGETING NEUROLOGICAL DISORDERS

ADDRESSING HUGE UNMET MEDICAL NEED WITH NO FDA-APPROVED THERAPIES

Clear Problem

- » TBI is the leading cause of **death and disability** among children and young adults in the U.S.
- » **> 69,000 TBI-related deaths** in the US in **2021**

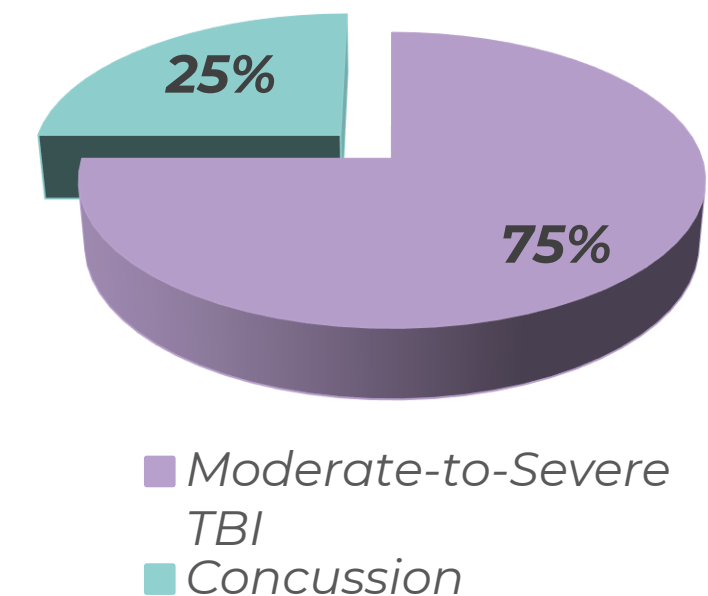
Large Market

- » Over **3 million** Americans sustain a TBI each year; annual medical costs **> \$48 billion**
- » Over **3.8 million sports- and recreation-related concussions** per year in the US*
- » **Epilepsy drug market** to reach over **\$5 billion** by **2024****

Great Opportunity

- » **No FDA-approved** therapies for TBI; estimated global market potential is **>\$4 billion**
- » Enormous markets for AD, PD and other indications that could be co-developed with big pharma partners

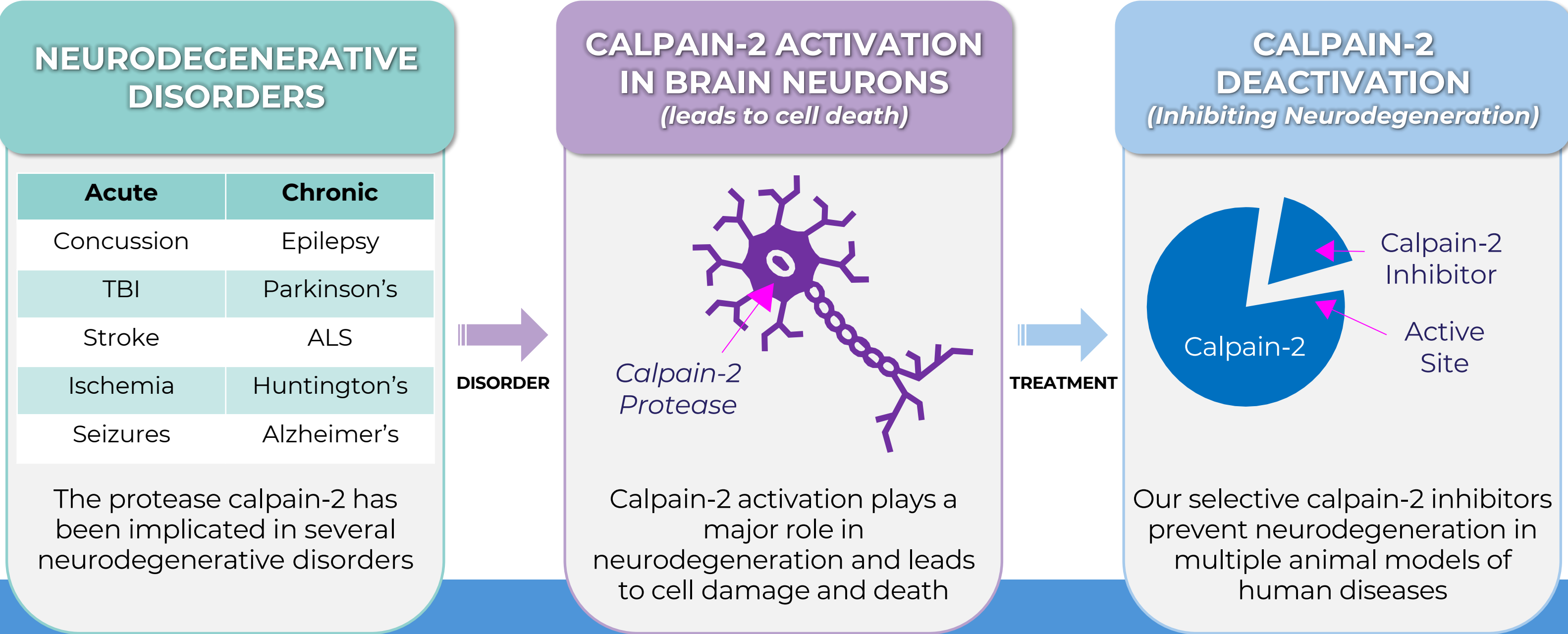
>3 Million Americans Sustain a TBI Annually



* Source: www.brainline.org
** Source: Grand View Research

OUR DISCOVERY

FOCUSING ON PREVENTING NEURONAL DAMAGE



NOVEL SCIENCE ON THE VERGE OF CLINICAL PROOF OF PRINCIPLE

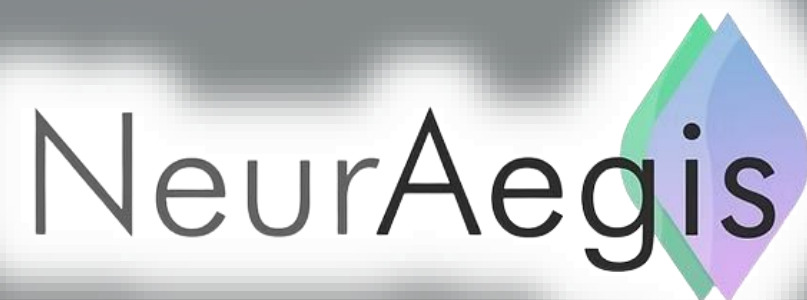
INITIAL TARGET INDICATIONS

CURRENTLY NO NEUROPROTECTIVE DRUGS IN THESE CONDITIONS

	Concussion	Status Epilepticus (Most extreme form of epilepsy with high mortality)	Glaucoma
Current Therapies	Currently no FDA approved medications	Currently no FDA approved medications	Currently no FDA approved medications
Overall Incidence	CDC estimates 3.8 million concussions occur in the U.S. annually	40 per 100,000 per year in the U.S. annually	CDC estimates 0.6% of the global population
Development Status	Lead candidate selected & Initiating IND- enabling program	Lead optimization & Demonstrated calpain-2 inhibition significantly limits neurodegeneration after seizure activity in animal model	Discovery
Current Funding	100% funded by DOD (through IND filing)	NINDS Award	Self Funded

CALPAIN-2 HAS BEEN IMPLICATED AND IS STRONGLY ACTIVATED IN EACH OF THESE DISORDERS

TBI/CONCUSSION SOLUTION



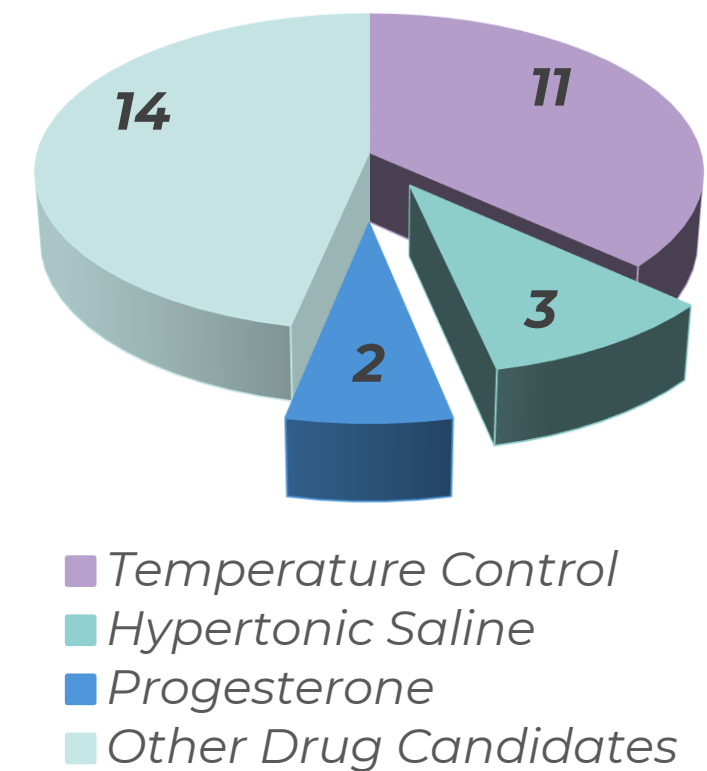
PREVIOUS CLINICAL TRIAL FAILURES

CLEAR UNDERSTANDING OF ALL CHALLENGES

Causes of the failures of all traumatic brain injury trials since 1993:

- » Unknown mechanism for neuronal damage
- » Lack of quantitative and predictive biomarkers to guide trials
- » Subjective outcome measures
- » Patient recruitment challenges

Reasons why TBI Clinical Trials failed

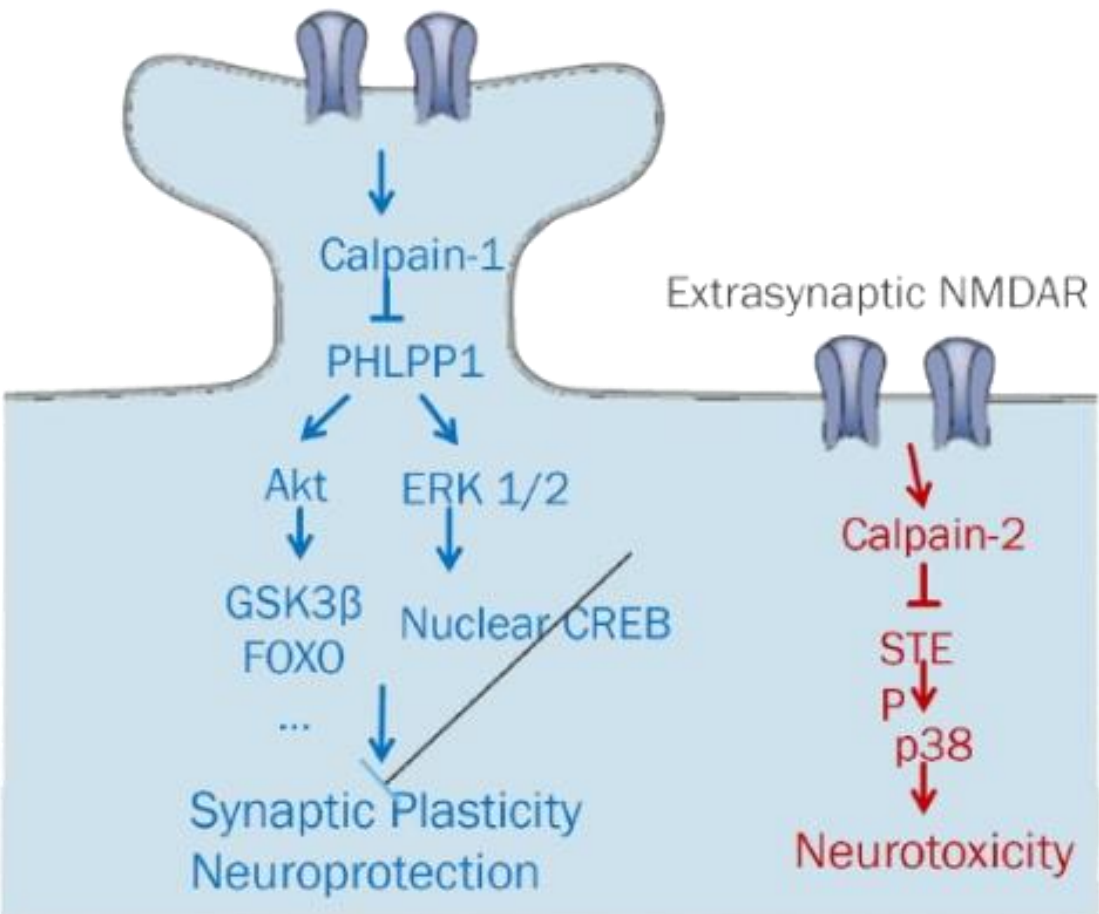


WE ARE ADDRESSING EACH OF THESE CHALLENGES IN OUR CLINICAL PLAN WITH THE GOAL OF OBTAINING THE FIRST FDA APPROVED THERAPY FOR CONCUSSION

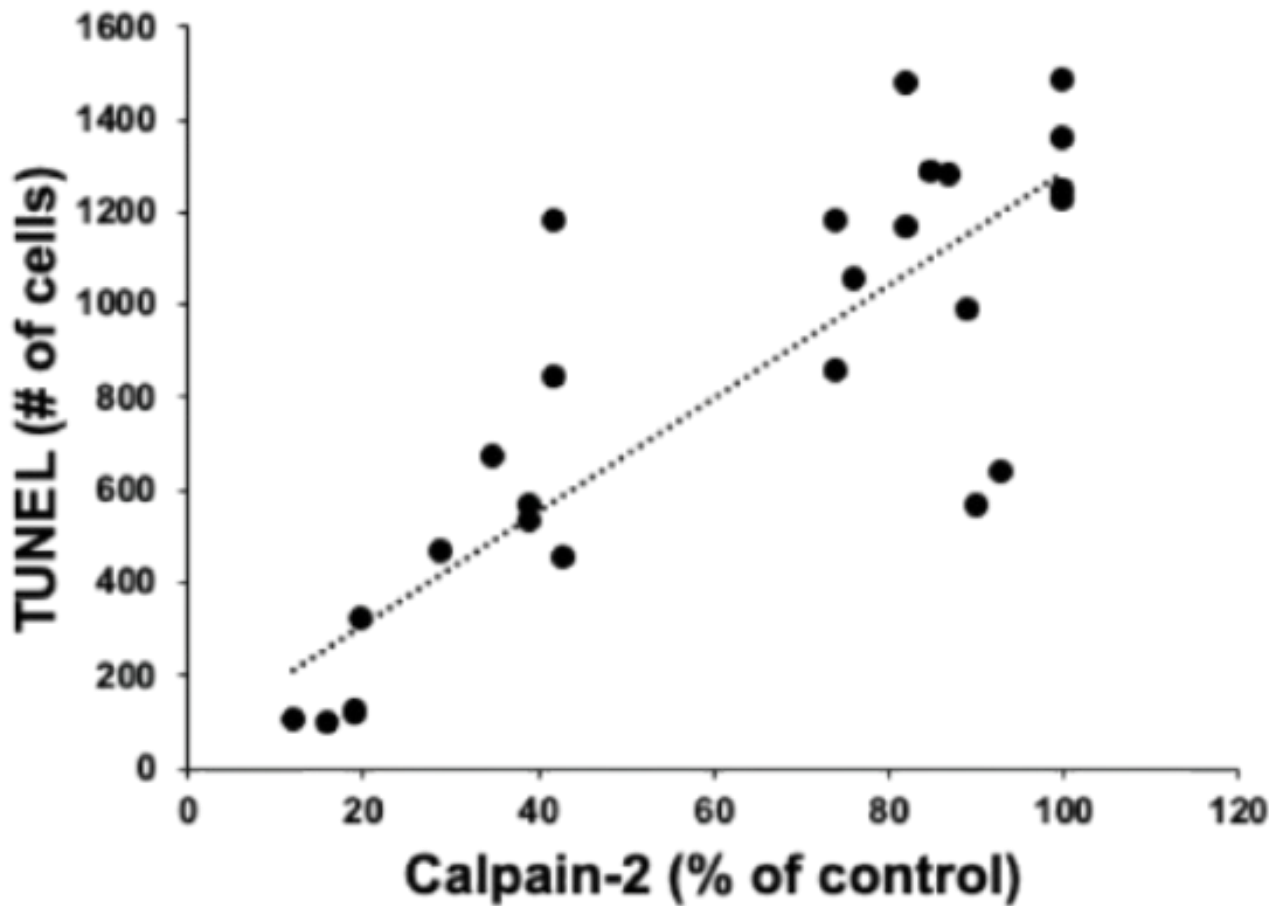
OUR PATH TO PREVENTING NEURONAL DAMAGE AFTER CONCUSSION

Our Key Discoveries	Initial Validation of our Solution
Identified a critical mechanism of neuronal damage and death resulting from concussion	We demonstrated that calpain-2 activation is correlated with cell damage and death following concussion

Calpain-1 and Calpain-2 play opposite functions in plasticity and neurodegeneration



Correlation of Cell Death and Calpain-2 Activity in Rat and Mouse Brain After TBI



INHIBITOR

OUR PATH TO PREVENTING NEURONAL DAMAGE AFTER CONCUSSION

Our Key Discoveries

Discovered an ***inhibitor*** of this neurodegenerative process

Initial Validation of our Solution

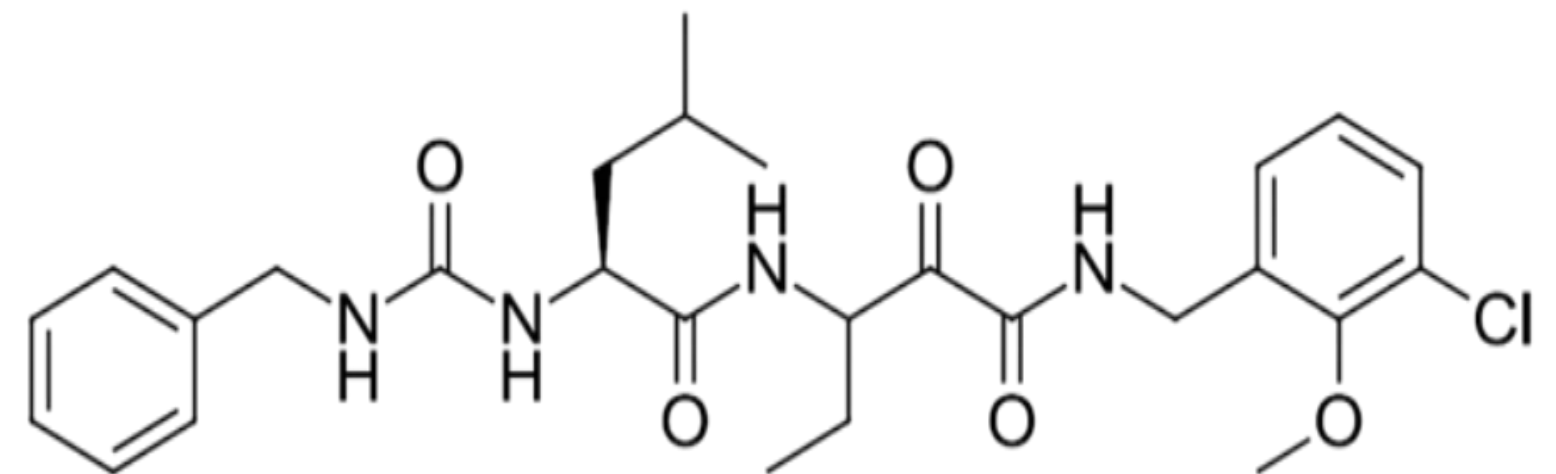
Identified novel calpain-2 inhibitors that provide significant protection when administered after concussion (animal models)

Calpain-2 inhibitor effects

Selective and potent calpain-2 inhibitor has **2 effects**:

- » Enhances synaptic plasticity/ Learning and Memory
- » Prevents neurodegeneration

NA-184 structure



NA-184 is very potent and selective for Calpain-2 when assayed against Human Calpains *In Situ*

NA184 was tested on calpain activity in homogenates of HEK cells expressing either calpain-1 or calpain-2, using spectrin degradation as the read-out

IC₅₀ for human calpain-1: no inhibition up to 10µM

IC₅₀ for human calpain-2: 1.2 nM

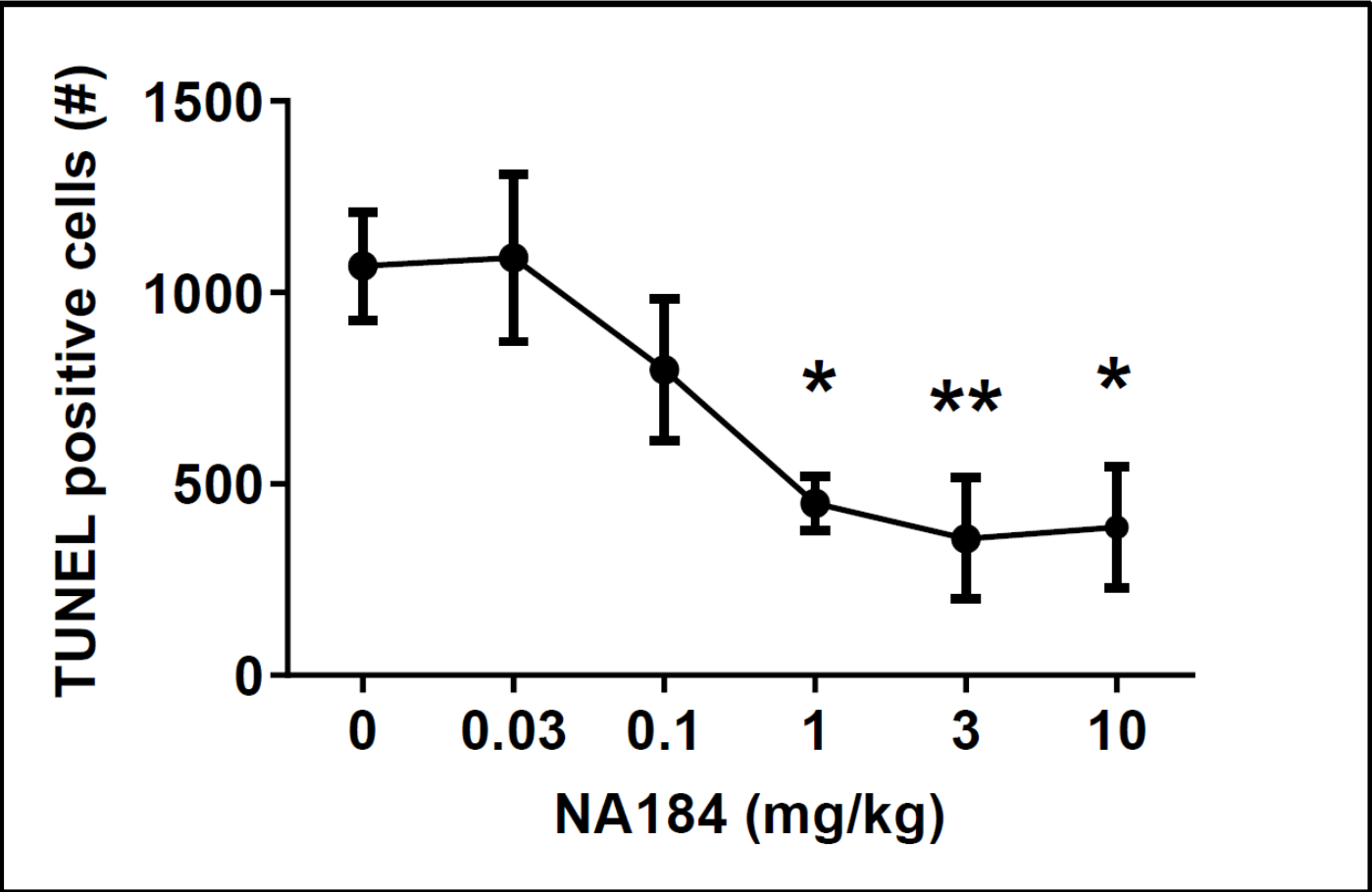
Ratio of selectivity: >3,000 fold

The cellular homogenate is closer to the in vivo situation as it contains the endogenous substrates of the two enzymes

NA-184 SIGNIFICANTLY REDUCES CELL DEATH & CALPAIN 2 LEVEL AFTER TBI IN MICE

Activity & Target engagement

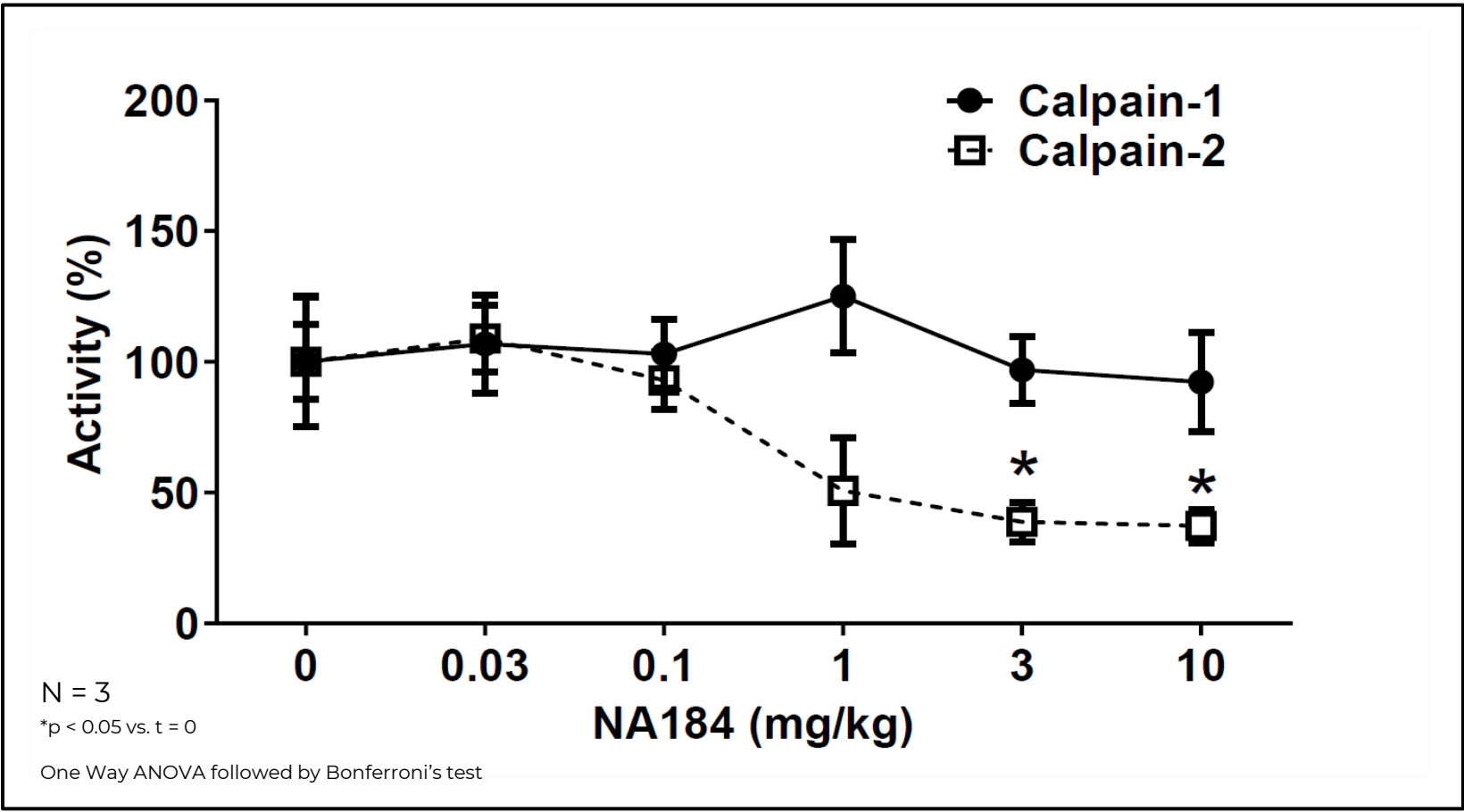
Intraperitoneal (IP) Injection of NA-184 1 h After TBI Significantly Reduces Cell Death 24 h After TBI in Male Mice



N = 6 for 0.1 mg/kg; N = 3 for other groups. *p < 0.05 and *p<0.01. One Way ANOVA followed by Bonferroni's test

ED₅₀ is approx. 0.13 mg/kg

IP injection of NA-184 1 h After TBI Reduces Calpain-2 But Not Calpain-1 Activation in Brain 24 h After TBI in Male Mice



N = 3

*p < 0.05 vs. t = 0

One Way ANOVA followed by Bonferroni's test

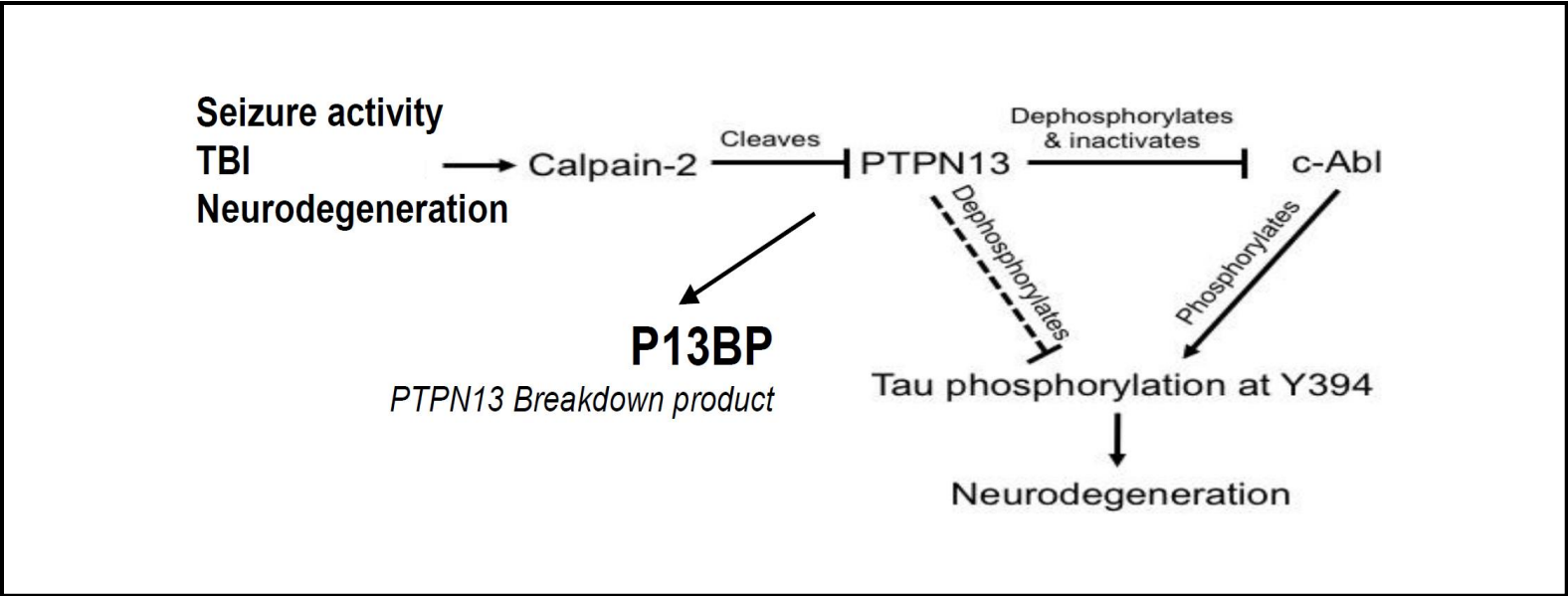
Cerebellar P2 fraction collected 24 h after TBI was used for measuring inhibitory activity of compounds against mouse endogenous calpain-1/-2

- Calpain-1 activity = calpain activity with 20 μ M Ca^{2+} - calpain activity with no Ca^{2+}
- Calpain-2 activity = calpain activity with 2 mM Ca^{2+} - calpain activity with 20 μ M Ca^{2+}

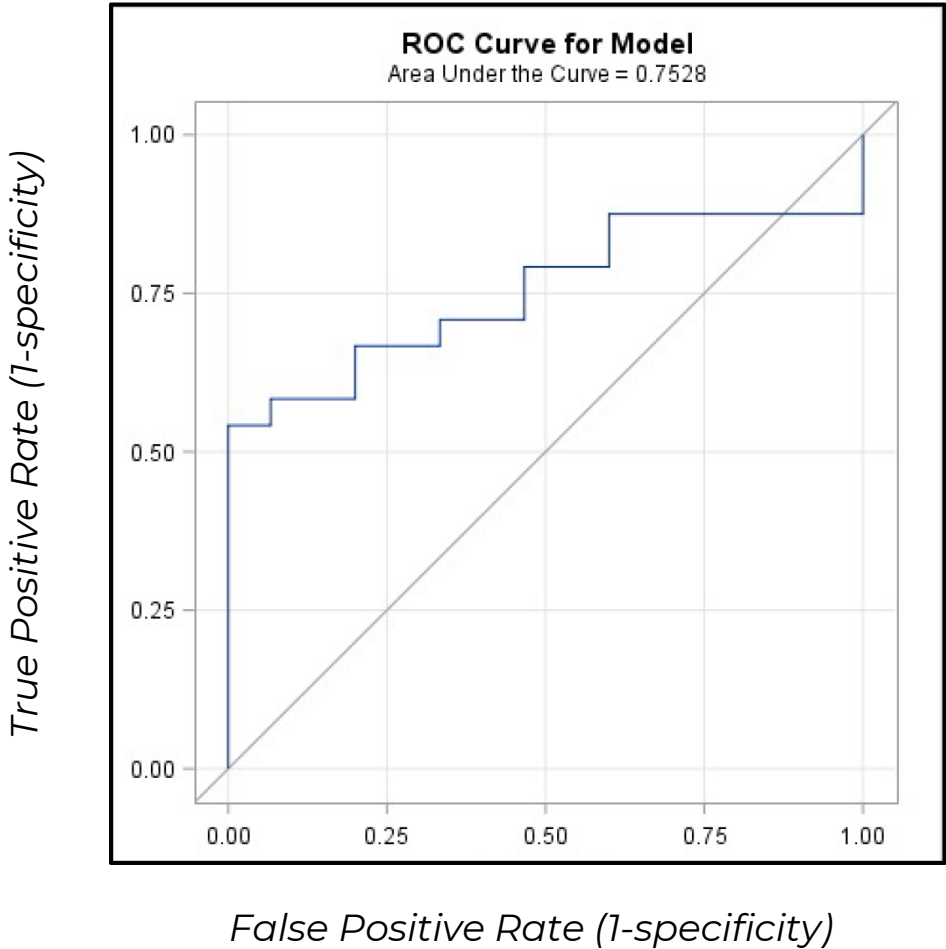
OUR PATH TO PREVENTING NEURONAL DAMAGE AFTER CONCUSSION

Our Key Discoveries	Initial Validation of our Solution
We identified blood-based biomarkers that directly correlates with cell death and drug target engagement	Discovered proprietary plasma biomarker (P13BP), which reflects calpain-2 activation in the brain (using human clinical samples in studies)

P13BP as a Blood Biomarker for TBI



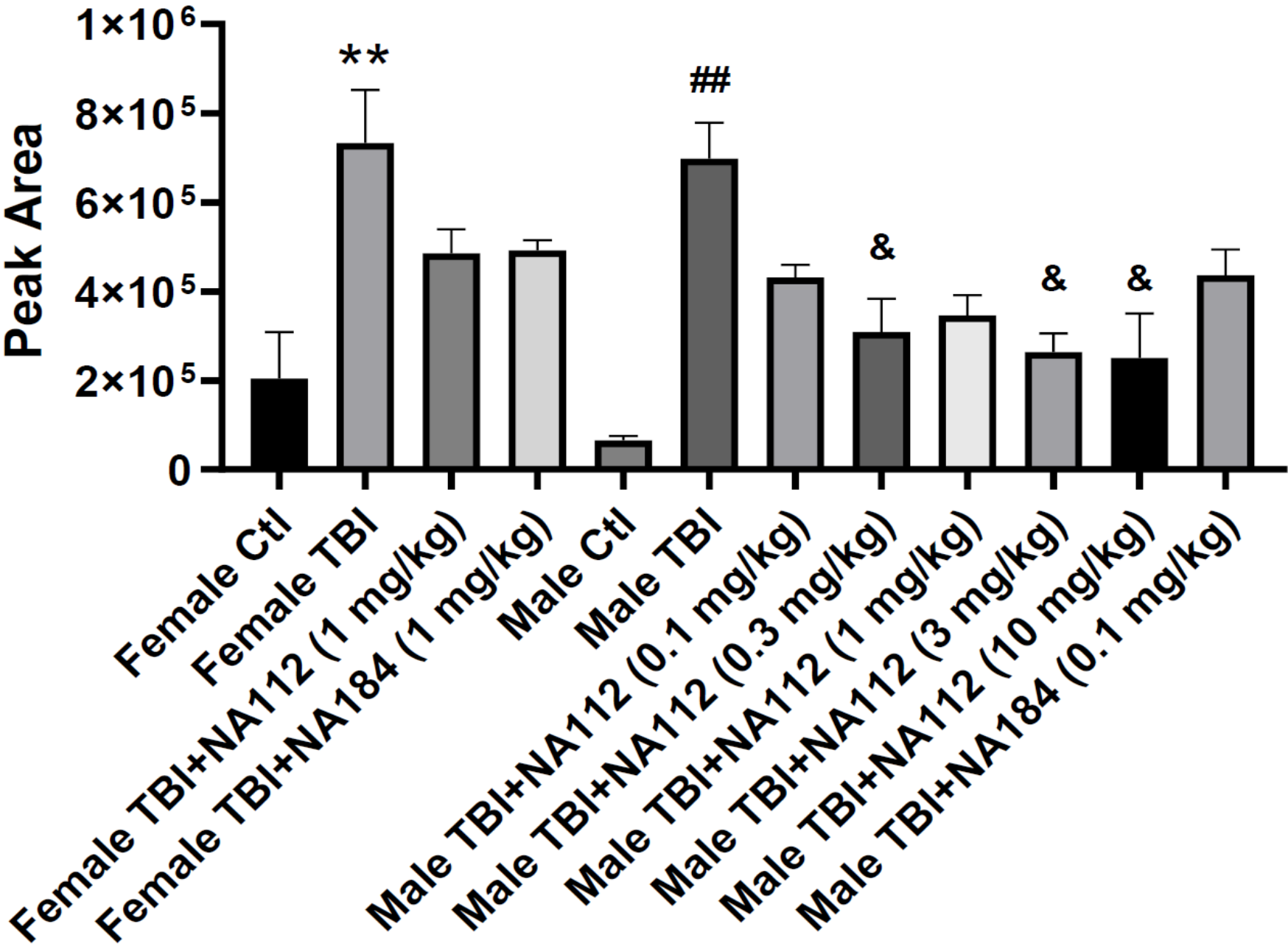
Receiver Operative Characteristics (ROC) Curve For Human Plasma P13BP Levels



P13BP plasma level is a good predictor of TBI severity in humans

PREVENTS TBI-INDUCED INCREASE IN PLASMA P13BP

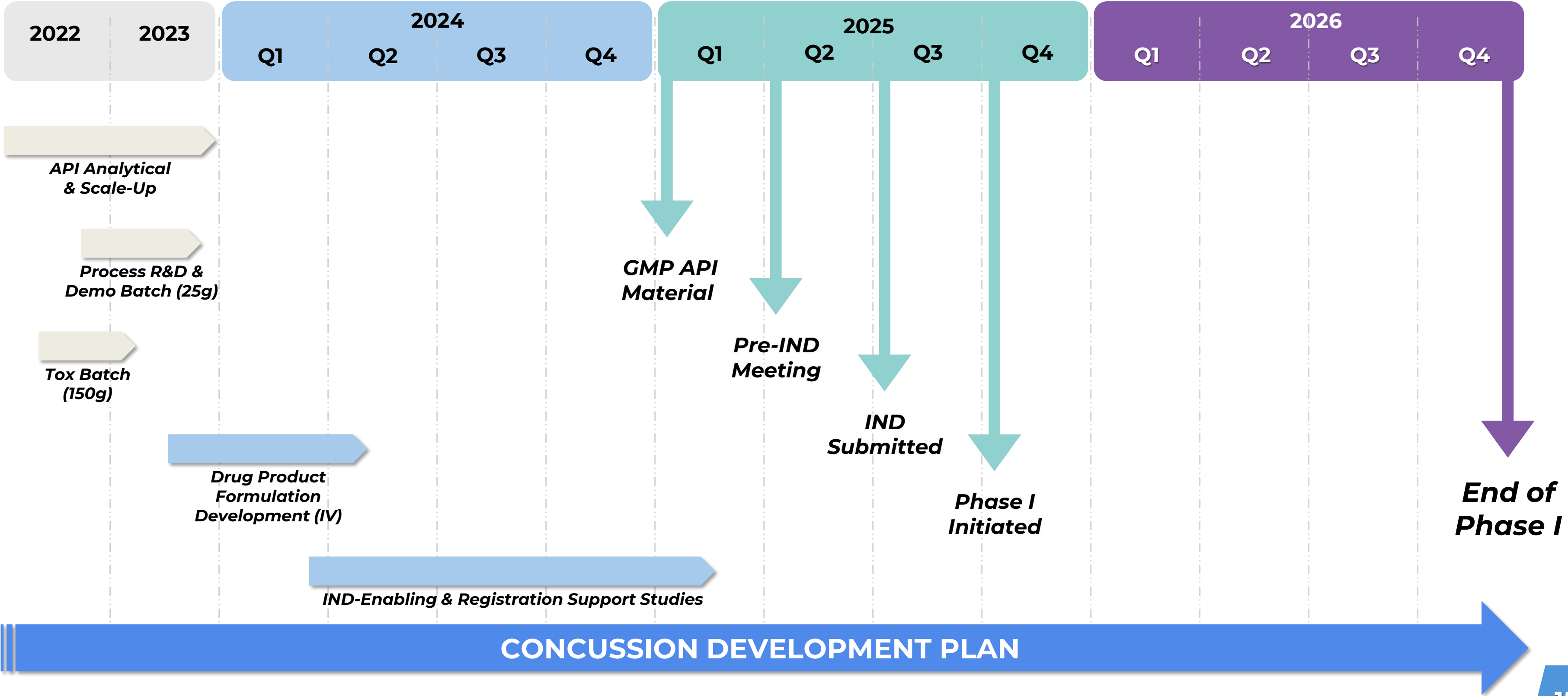
P13BP LEVELS ARE SIGNIFICANTLY INCREASED IN THE PLASMA OF MALE AND FEMALE MICE 24 H AFTER TBI; IP INJECTION OF NA-112 OR NA-184 REDUCES THE INCREASE IN P13BP AFTER TBI



NA-184 prevents TBI-induced increase in plasma P13BP

** : p < 0.01 vs Control females ##: p < .01 vs Control males &: p < 0.05 vs Control males

CONCUSSION DEVELOPMENT PROGRAM (2022-2026)



NEURAEGIS LEADERSHIP



NEURAEGIS LEADERSHIP TEAM

PROJECT LED BY RENOWNED EXPERTS IN NEUROSCIENCE

Top Management



Michel Baudry, PhD

Founder & Chief Scientific Advisor

World-renowned neuroscientist with over 400 publications



Stella M. Sung, Ph.D.

President & CEO

C-Level operating executive & VC, helped build multiple life science firms



Michael G. Palfreyman, PhD, DSc

Drug Development Advisor

Expert in preclinical drug development

Key Managers



Philippe Bey, Ph.D.

Chemistry

*ArQule, Sanofi Aventis, Hoechst Marion Roussel
Ph.D. from University Louis Pasteur (chemistry)*



Greg Coulter, Ph.D.

Biological Chemistry

*University of Guelph, Canada
President, CTM Solutions, LLC
Pharmaceutical Development Specialist*



Kathy Fosnaugh, Ph.D.

Program Management

*Aminex Therapeutics, Sirna Therapeutics,
Ligand Pharmaceuticals
Ph.D. from Cornell University (microbiology)*



NEURAEGIS LEADERSHIP TEAM

PROJECT LED BY RENOWNED EXPERTS IN NEUROSCIENCE



Top Management



Michel Baudry, PhD
Founder & Chief Scientific Advisor

World-renowned neuroscientist with over 400 publications

- » World-renowned neuroscientist, life science entrepreneur and professor. Co-Founder of 5 biotechnology companies across two countries.
- » Identified calpain-2's role in neurodegeneration and discovered novel calpain-2 protease inhibitors. Earned his Ph.D. (Biochemistry) from University Paris VII (Paris, France)



Stella M. Sung, Ph.D.
President & CEO

C-Level operating executive & VC, helped build multiple life science firms

- » C-level (CEO/CBO) life Science/biotech executive with 25 years of experience in both operating and venture capital roles.
- » Successful track record in negotiating, establishing, and executing transactions, agreements, and partnerships.
- » Extensive experience in raising capital, building companies; served on or chaired Boards of >10 life science companies. Earned her Ph.D. (Chemistry) from Harvard University (National Science Foundation Pre-Doctoral Fellow).



Michael G. Palfreyman, PhD, DSc
Drug Development Advisor

Expert in preclinical drug development

- » Scientific leader in pharmaceutical industry for over 40 years.
- » Internationally recognized leader in neuropharmacology and drug development and clinical Proof of Concept.
- » Successfully led multiple IND filings and clinical programs.
- » Holds a D.Sc. and a Ph.D. degree in Pharmacology, both from the University of Nottingham, UK.



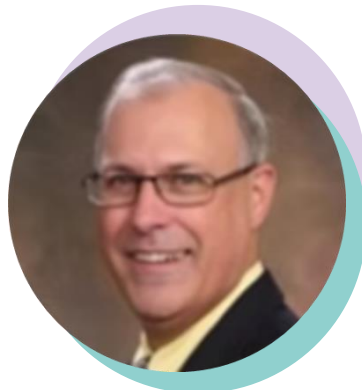
Philippe Bey, Ph.D.
Chemistry

ArQule, Sanofi Aventis, Hoechst Marion Roussel
Ph.D. from University Louis Pasteur (chemistry)

- » President of Bey Consulting, Inc. Participated in and led teams responsible for over 30 INDs across several therapeutic areas, including oncology.
- » Oversaw bulk drug supply for preclinical and clinical studies., Post-doctoral training at the California Institute of Technology.
- » Adjunct professor of medicinal chemistry at the University Louis Pasteur for 12 years.
- » Authored nearly 100 publications, 21 book chapters, and holds 63 issued patents. Inventor of difluoromethylornithine (DFMO), the standard of care for African sleeping sickness.



Key Managers



Greg Coulter, Ph.D.
Biological Chemistry

University of Guelph, Canada
President, CTM Solutions, LLC
Pharmaceutical Development Specialist

- » A consultancy specializing in CMC and clinical trial materials.
- » Over 20 years in GLP, GMP, and GCP regulated drug product development. Specializes in drug substance synthesis, scale-up, process development, analytical characterization, and regulatory CMC guidance.
- » Postdoctoral training at the University of Denver. Previous Roles: Held positions at Synthetech, Cell Therapeutics, and SNBL USA, with increasing responsibilities. Recent Work: Contributed to the FDA approval of Contrave (weight loss drug) and Zydelig (PI3K delta inhibitor).



Kathy Fosnaugh, Ph.D.
Program Management

Aminex Therapeutics, Sirna Therapeutics, Ligand Pharmaceuticals
Ph.D. from Cornell University (microbiology)

- » Biotech consultant specializing in preclinical and Phase 1 clinical program management and drug development at KL Fosnaugh Consulting, LLC
- » Over 30 years of experience in translational research and development of small molecule and nucleic acid therapeutics.
- » Has served as Program Manager for three IND submissions and Phase 1 clinical trials.
- » Holds a BS in microbiology from the University of Illinois and a PhD in microbiology from Cornell University.
- » Completed postdoctoral research at the University of California, San Diego. Certified in Biomedical Regulatory Affairs from the University of Washington



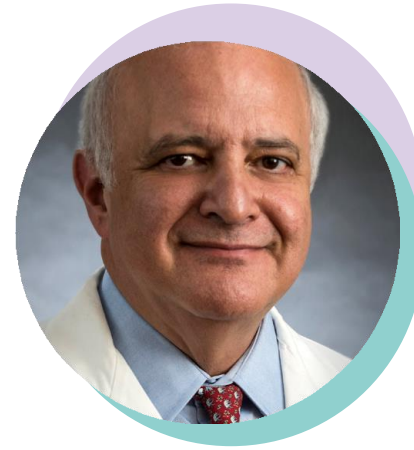
SCIENTIFIC & CLINICAL ADVISORY BOARD

PROJECT SUPPORTED BY WORLWIDE RECOGNIZED PROFESSIONALS



Dr. Imad Najm, MD
Director of the Epilepsy Center
Cleveland Clinic

- » Director of the Epilepsy and Clinical Neurophysiology Fellowship Program at Cleveland Clinic (2000-2003).
- » Head of the Section of Adult Epilepsy at Cleveland Clinic (2003).
- » Medical Doctor degree from Saint Joseph University, Beirut, Lebanon (1988).
- » Post-Doctoral Research Fellow in Neurobiology at the University of Southern California (1989-1992), focused on mechanisms of status epilepticus-induced neuronal injury.
- » Certified in adult neurology by the American Board of Psychiatry and Neurology (1998). Added qualification in clinical neurophysiology (1999).



Dr. Ramon Diaz-Arrastia, MD, PhD
Director, Traumatic Brain Injury Clinical Research
Center The University of Pennsylvania
Perelman School of Medicine

- » Attending Neurologist at the Hospital of the University of Pennsylvania and Penn Presbyterian Medical Center
- » His research focuses on the molecular mechanisms of neuronal injury, neurodegeneration, traumatic brain injury, and epilepsy.
- » Professor of Neurology at the Uniformed Services University of Health Sciences and Investigator at the National Institute of Neurological Disorders and Stroke. (2011-2016)
- » Adjunct Professor at the University of Texas at Dallas.(2004)
- » MD from Baylor College of Medicine (1988)
- » Authored numerous articles in peer-reviewed journals, primarily focused on neurodegeneration, traumatic brain injury, and post-traumatic epilepsy.



Barry Jordan, MD, MPH
Chief Medical Officer
Rancho Los Amigos Nat Rehab Center

- » Assistant Medical Director and Interim Chief Medical Officer at Burke Rehabilitation Hospital, White Plains, NY.
- » Team physician for U.S.A. Boxing.
- » Chief Medical Officer for the New York State Athletic Commission (20 years).
- » B.A. in Neurophysiology from the University of Pennsylvania.
- » MD from Harvard Medical School.
- » MPH from Columbia University

BUSINESS ADVISORS

PROJECT SUPPORTED BY EXPERIENCED EXPERTS IN STRATEGY & MANAGEMENT



Lewis Geffen

Corporate Counsel; Co-Chair Life Science Practice, Mintz Levin

- » A corporate and licensing attorney focused on the life sciences, with decades of experience helping companies in the sector tackle strategic issues and complete complicated transactions.
- » He is the go-to outside lawyer for preeminent biotechnology, pharmaceutical, medical device, and medtech companies, shepherding them through financing and licensing deals that fund drug development.
- » Active member of the Biotechnology Industry Organization, Massachusetts Biotechnology Council, BioCOM, AdvaMED, MassMEDIC and other professional organizations. He is also a member of the American and Boston Bar Associations



Peter Corless

IP Counsel; Partner, Fox Rothschild LLP

- » Strategic adviser on patent and trademark portfolio management and intellectual property due diligence for businesses, academic institutions, and investors in North America, Europe, and Asia.
- » Works with academic and medical research institutions, biotech, pharmaceutical, manufacturing companies, venture capital firms, underwriters, and medical technology companies.
- » Former partner at a Boston-based international law firm and co-chair of its patent procurement practice group.
- » Graduate studies in chemistry and worked as a synthetic organic chemist for E.I. DuPont de Nemours Co. (now part of Bristol-Myers Squibb).



NEURAEGIS COMPETITIVE ADVANTAGE



COMPETITIVE ADVANTAGE

NEURAEIGIS TAKES A DIFFERENTIATED APPROACH

NeurAegis's selective calpain-2 inhibitor, NA184, is differentiable from current TBI treatments as well as other products in development:

Current TBI treatment options include psychostimulants, antidepressants, antiparkinsonian agents, and anticonvulsants

PNT001 (Pinteon)

Tau antibody

AST-004 (Astrocyte Pharm)

Adenosine receptor agonist

N-Acetyl-cysteine (Neuronasal)

Antioxidant

OXE-103 (Oxeia)

Ghrelin

VAS203 (Vasopharm)

Inhibitor of nitric oxide synthase

NeuroSTAT (NeuroVive Pharmaceutical)

Proprietary formulation of cyclosporin; targeting mitochondria

CEVA101 (Cellvaton)

Autologous bone marrow-derived mononuclear cells (bmmncs)

FAVORABLE MARKET DYNAMICS

» TBI drugs in pipeline with novel MoA are still **years from commercialization** - expected in the 2024-2026 timeframe

» Potential for **relatively short development** and regulatory timelines

» **Blockbuster potential** for first in class or best in class therapeutics, and room for multiple **"winners"**

» In 2019, **NeuroSTAT** received **Fast Track designation** from the FDA

» **VAS203** expected to reach peak revenues of **\$2.5B by 2030**

SERIES A FINANCING



PROPOSED TERMS FOR SERIES A RAISE

HIGHLY PROMISING INVESTMENT FOR NEW ENTRANTS

» **\$10M Series A Preferred financing**

- Funds to key milestones/value inflection points (Ph 1 in concussion)
- 2 years operating runway

» **Fully diluted pre-money valuation= \$15M**

» **Liquidation preference**

- First pay one times the Original Purchase Price plus declared and unpaid dividends on each share of Series A Preferred; (or, if greater, the amount that the Seed Preferred would receive on an as-converted basis). The balance of any proceeds shall be distributed pro rata to holders of Common Stock.

» **Weighted average anti-dilution protection**

» **One Board or Observer seat** (if desired) for Series A Preferred investor class

COMPARABLES

ACTIVE ACQUISITION MARKET IN THE NEURAL FIELD



- » A clinical-stage biopharmaceutical company that combines a deep understanding of the biology and neurocircuitry of the brain?
- » Focused on the discovery, development and commercialization of innovative new treatments for the neurodegenerative community PD, AD, Movement disorder.

» *Acquired By: Abbvie Aug 2024, \$8.7 Billion*



- » Is working on innovative solutions for Parkinson's disease treatment
- » NeuroDerm's flagship product ND0612 is currently in late stage development for Parkinson's disease patients experiencing motor fluctuations.

» *Acquired By: Mitsubish Tanabe Pharma, 2017 \$1.1 Billion*



- » Intravenous formulation for acute central nervous system conditions such as stroke (ischemic and hemorrhagic), traumatic brain injury.

» *Acquired By: Biogen May 15, 2017 \$120M*

COMPARABLES

AVERAGE MARKET CAP AT THE TIME OF IPO IS \$570 MILLION

Compagnies involved in developing treatments for neurological disorders and have gone public in the last few years. Their average market capitalization at the time of IPO is \$570 million.

<i>Company Name</i>	<i>Stock Ticker</i>	<i>IPO Date</i>	<i>Market Capital at IPO (USD)</i>
Amylyx Pharmaceuticals	NASDAQ:AMLX	Jan 7, 2022	1,069,196,253
Longboard Pharmaceuticals	NASDAQ:LBPH	Mar 12, 2021	270,671,840
Foghorn Therapeutics	NASDAQ:FHTX	Oct 23, 2020	571,116,832
Ovid Therapeutics	NASDAQ:OVID	May 5, 2017	369,029,055

SUMMARY OF INVESTMENT OPPORTUNITY VALIDATED KPIs FOR INVESTORS' ANALYSIS

NeurAegis's therapeutics have the potential to transform treatment of TBI and other neurological disorders and to achieve blockbuster status (\$1+ billion in annual revenues)

» Novel therapeutics with **blockbuster potential**

- **First-in-class** to prevent neuronal death
- Addressing **multiple neurological disorders**, large markets
- **No current FDA-approved medication**—huge unmet medical need!
- **Plasma biomarker**

» Strong **Intellectual Property**

» Experienced **Leadership Team**

» Reduced **clinical risk**

» Significant Upside **Potential**

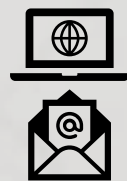
- **Key milestones achieved** on grant funding (validative and non dilutive)
- First **institutional round**
- Exit comparables suggest **10x return potential**

For further details

Please contact



Stella M. Sung, Ph.D.
President & CEO



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[WWW.NEURAEGIS.COM](http://www.Neuraegis.com)

Serie A Fundraising - Investors Deck

APPENDIX





Western University
OF HEALTH SCIENCES

» Exclusive license with Western University of Health Sciences

» Multiple pending patent applications

- Composition of matter and therapeutic uses of calpain inhibitors
- Novel blood biomarker for diagnosis and monitoring of acute neuronal damage
- Potential process patents for lead compounds

» Received favorable International Search Report and Written Opinion for composition of matter claims

KEY MILESTONES

Accomplished

2016 - 2017

- Company formation
- Identified calpain-2's role in neurodegeneration

2018 - 2019

- Discovered novel calpain-2 protease inhibitors (NCE's)
- Demonstrated efficacy in several preclinical animal models
- Obtained \$3.8M DOD grant

2020

- Selected clinical candidate for concussion (NA184)
- Identified and screened novel compounds for additional indications

2021 - 2023

- Initiate IND-enabling program for concussion/TBI

2023

- Awarded >\$1M in additional grant funding, including NIH grant for status epilepticus

Up Next

2024

- Complete pre-IND studies for concussion
- Secure funding for first-in-human clinical studies (concussion) and earlier stage programs

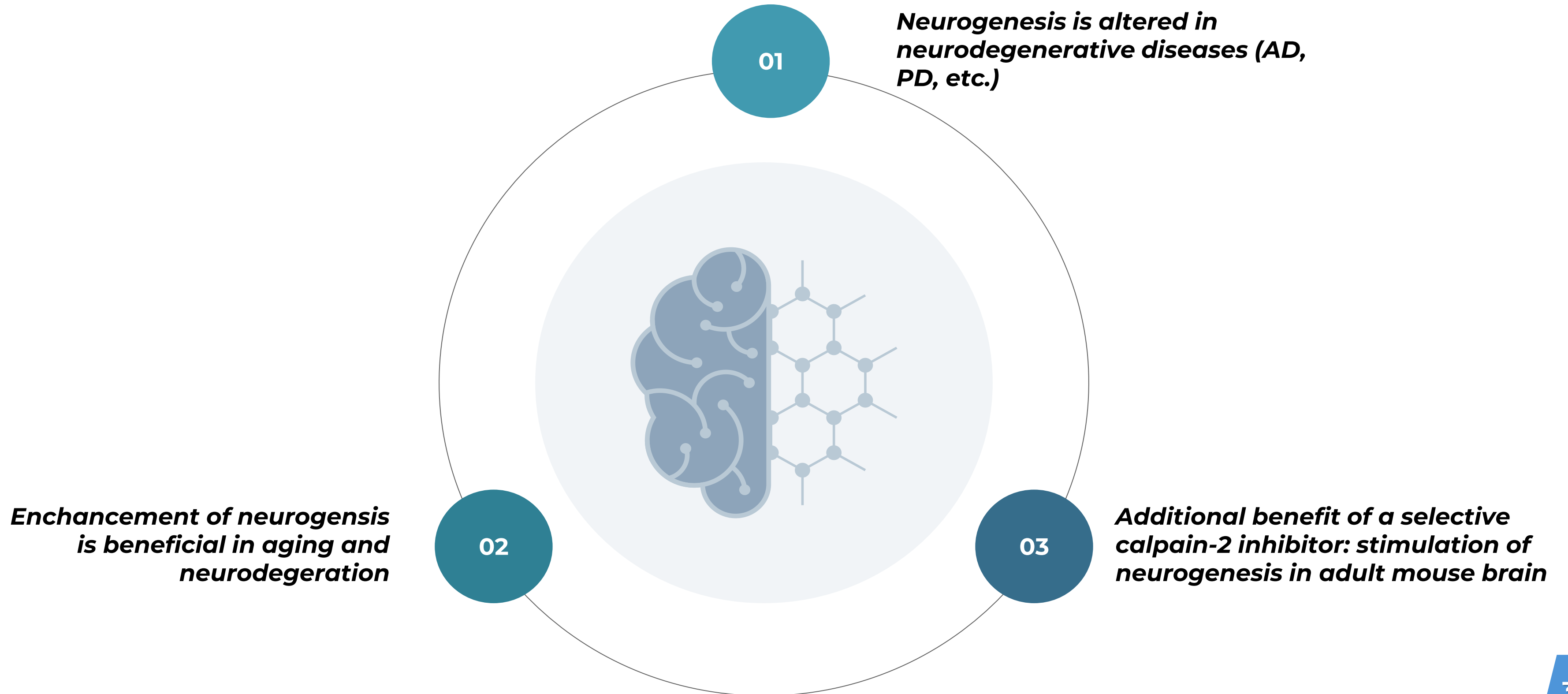
2024 - 2025

- Select clinical candidate for status epilepticus (most severe form of epilepsy)
- Initiate Phase 1 studies for concussion
- Initiate IND-enabling program for status epilepticus

2026

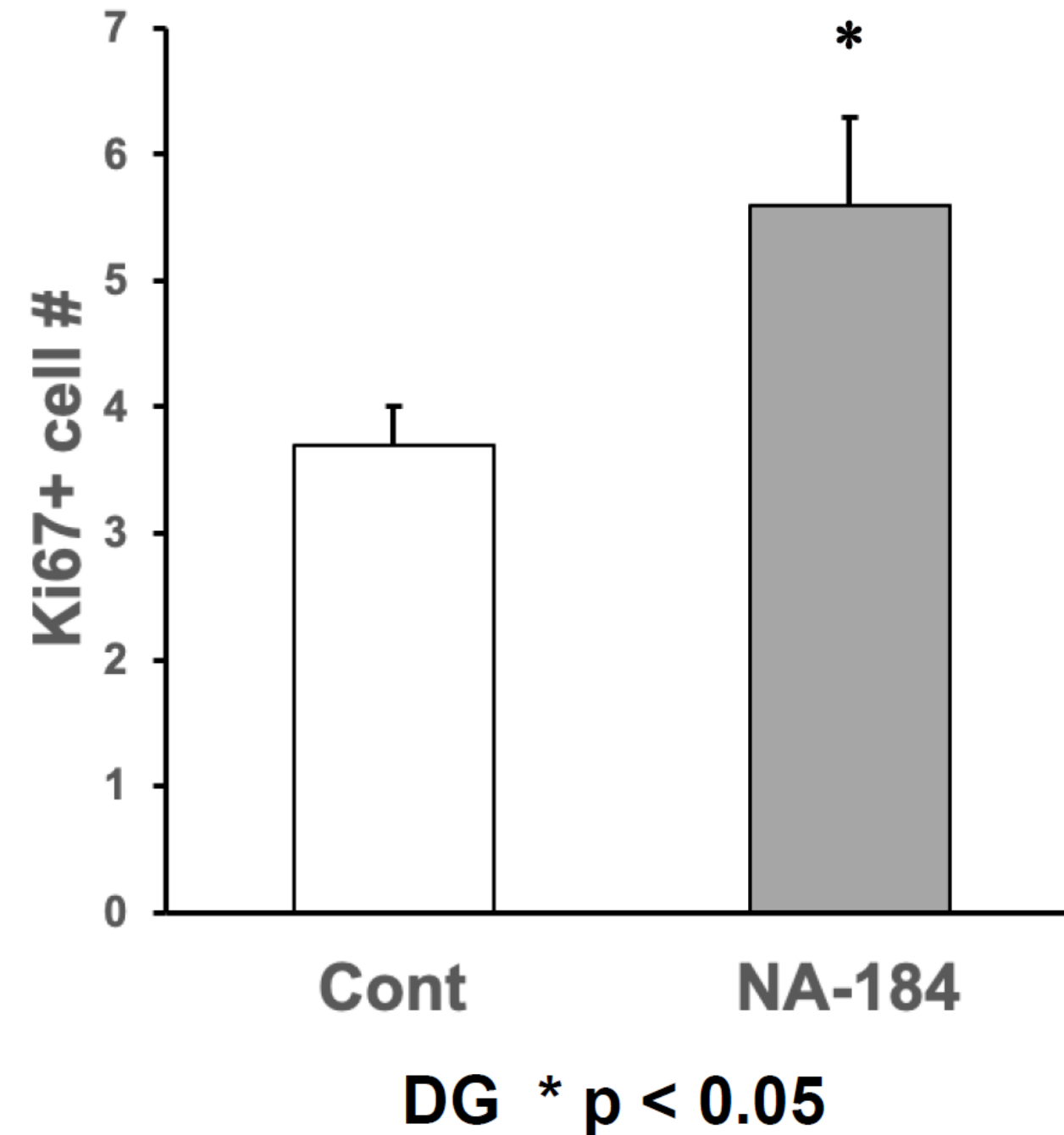
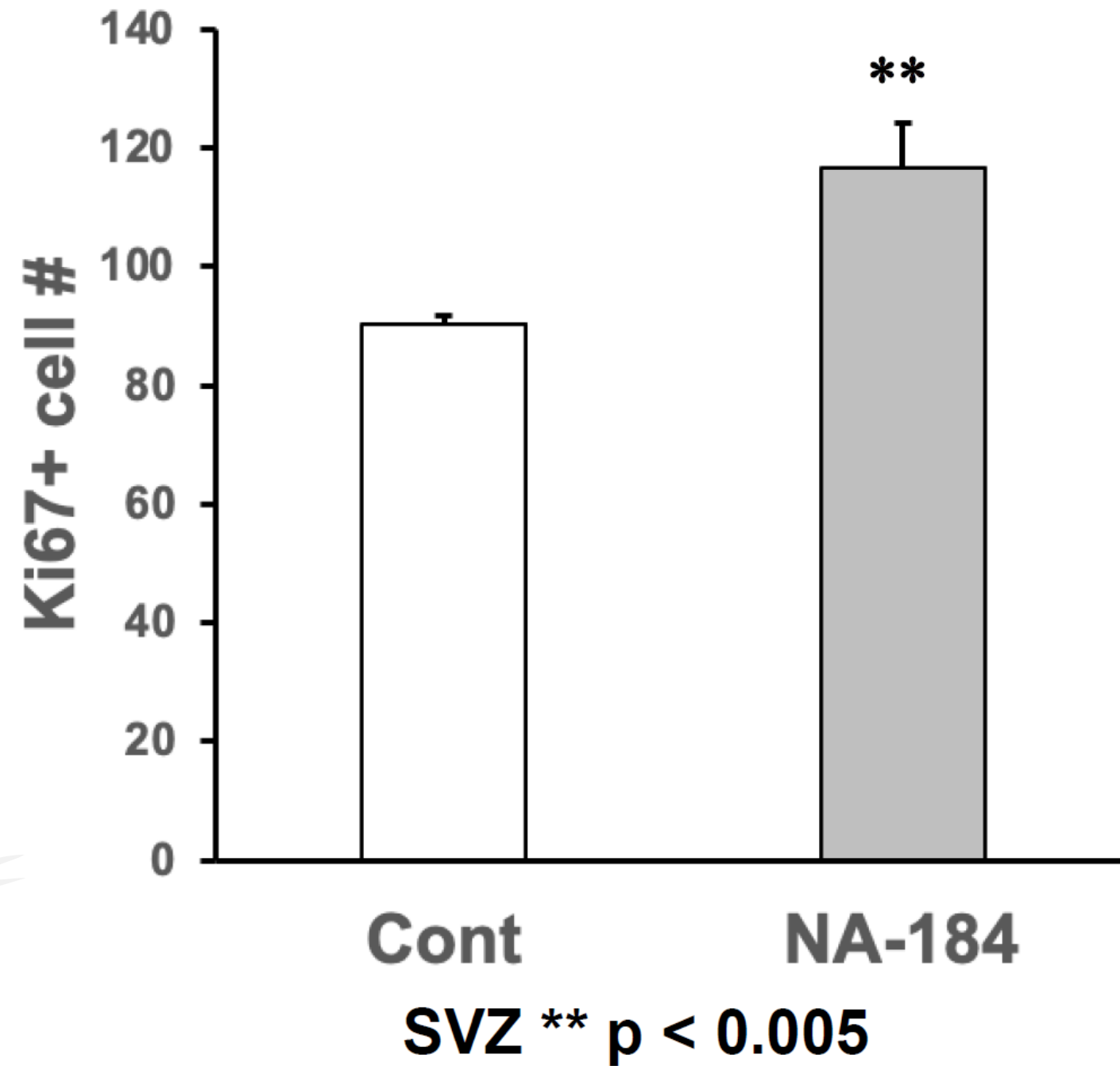
- End of Phase 1

STIMULATION OF NEUROGENESIS AS A POTENTIAL TREATMENT FOR NEURODEGENERATIVE DISEASES

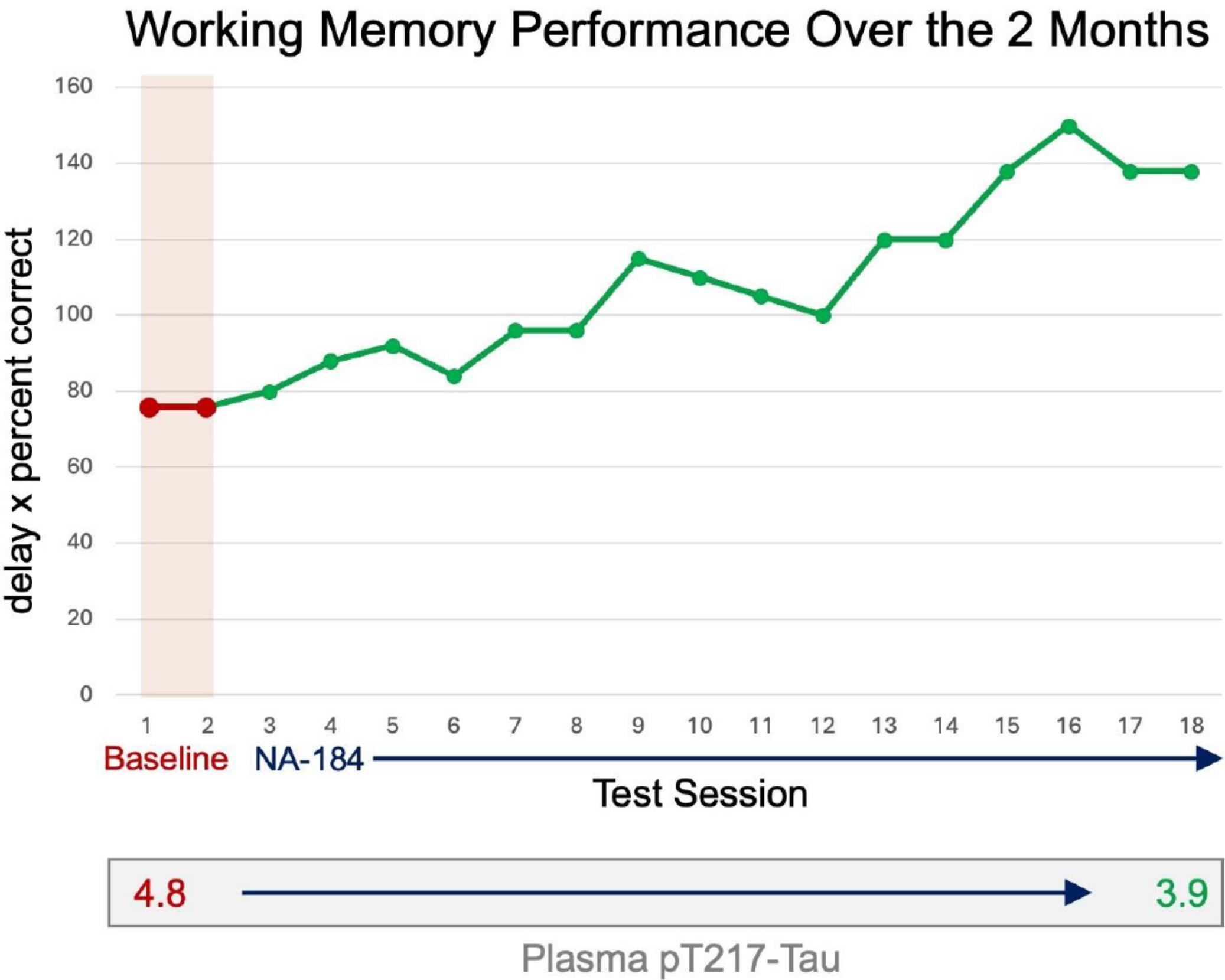


INHIBITOR: NA-184

STIMULATES NEUROGENESIS



DAILY INJECTIONS OF NA-184 FOR 2 MONTHS IN AGED MONKEYS INCREASE LEARNING AND MEMORY AND DECREASE PLASMA P-TAU



REFERENCES / PUBLICATIONS

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- Baudry, M. and Bi, X. Calpain-1 and calpain-2: the yin and yang of synaptic plasticity and neurodegeneration. *Trends in Neurosciences* 2016 Feb 10. **39**: 235-245. pii: S0166-2236(16)00020-5. doi: 10.1016/j.tins.2016.01.007, 2016.
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- Wang Y, Hersheson J, Lopez D, Ben Hamad M, Liu Y, Lee K-H, Pinto V, Seinfeld J, Wiethoff S, Sun J, Amouri R, Hentati F, Baudry N, Tran J, Singleton AB, Coutelier M, Brice A, Stevanin G, Durr A, Bi X, Houlden H and Michel Baudry M. Defects in the *CAPN1* gene result in alterations in cerebellar development and in cerebellar ataxia in mice and humans. *Cell Reports* **16**: 79-91, 2016.
- Seinfeld, J., Baudry, N., Xu, X., Bi, X. and Baudry, M. Differential activation of calpain-1 and calpain-2 following kainate-induced seizure activity in rats and mice. *eNeuro* Sep 6;3(4). pii: ENEURO.0088-15.2016. doi: 10.1523/ENEURO.0088-15, 2016.
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- Wang, Y., Hall, R.A., Lee, M., Kamgar-parsi, A., Bi, X. and Baudry, M. The tyrosine phosphatase PTPN13/FAP-1 links calpain-2, TBI and tau phosphorylation. *Scientific Reports* | 7: 11771 | DOI:10.1038/s41598-017-12236-3. 2017.
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